

MATH 526 Lab 5 Fall 2006

Part I

In this part we learn to use the command `\` and the different display formats of Matlab.

The MATLAB command for solving $\mathbf{Ax} = \mathbf{b}$ is $\mathbf{A} \backslash \mathbf{b}$. The matrix $\mathbf{A}$ is factorized by Gaussian elimination using partial pivoting. Then the factors are used to find the solution $\mathbf{x}$.

Create the matrix $\mathbf{A}$ and the vector $\mathbf{b}$:

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & 5 & 7 \\ 2 & 4 & 6 & 8 \\ 1 & 0 & 5 & 7 \\ 1 & 3 & 7 & 4 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

Now type:

```
>>x=A\b
>>A*x
```

Matlab can display numerical results in several formats. Here we experiment with some of them.

```
>>format long
>>x=A\b
```

The components of the solution $\mathbf{x}$ have more digits than before. (The default format is **format short**).

```
>>format long e
>>x=A\b
```

The components of the solution $\mathbf{x}$ are now in the floating point format.

```
>>format rat
>>x=A\b
```

The components of the solution $\mathbf{x}$ are now rational numbers.

Use **help** to find out more about the command **format**.

Clear all variables before you work on Part II.